

Professional Vue.js

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In this 5-day course, students will start with the fundamental concepts and structures of Vue.js development, and quickly progress to learning about and solving problems with the most up-to-date techniques and tools in Vue.js development, including: Components, Routing, and Vuex.

Professional Vue.js is a complete introduction to modern front-end development. It covers the syntax, conventions, and best practices of modern JavaScript development while also teaching Vue.js. Through a series of hands-on exercises and demonstrations, students will learn how to develop and test complete Vue applications.

TARGET AUDIENCE

This course is designed for experienced front-end web developers who wish to begin using Vue.

OBJECTIVES

After taking this course, students will be able to:

- Install, configure, and use modern web tooling
- Learn to create test suites for Vue
- Write ES2018 code and compile it using Babel
- Understand what Vue.js is and what problem it solves
- Explore the basic architecture of a Vue.js application
- Gain a deep knowledge of Vue.js components
- Build a Single Page Application with Vue Router
- Use Vuex for maintaining state in a Vue.js application
- Use Local Storage with Vue
- Render Vue.js on the server
- Learn Vue.js best practices

TOPICS

- Modern JavaScript tooling
- Node.js
- Debugging JavaScript
- Object-oriented JavaScript
- A 10,000 foot view of Vue
- All the basic Vue capabilities and features
- Communication between components
- Data binding
- Communication with the server via Ajax
- State management with Vuex
- Creating SPAs with Vue
- Asynchronous JavaScript - promises, async/await (time permitting)
- Animation and Transitions in Vue (time permitting)

Audience

This course is designed for experienced front-end web developers who wish to begin using Vue.

Prerequisites

Before taking this course, students should have solid experience with HTML5, CSS3, and JavaScript

Duration

Five days

OUTLINE

- I. Introduction
- II. Development Ecosystem
 - a. Code Editors and IDEs
 - b. Lab 01: Installing and Configuring Visual Studio Code
- III. Vue Quickstart
 - a. Lab 02: Get Started with vue-cli
 - b. Introducing our Real World Project
 - c. Lab 03: Your First Component
 - d. Lab 04: Create More Components
 - e. Lab 05: Writing Unit Tests for Vue Components
- IV. Intro to NodeJS
 - a. What is Node?
 - b. A history of node
 - c. Lab 06: Getting Started with Node.js and npm
 - d. Lab 07: Initializing npm and Writing npm Scripts
 - e. Lab 08: Writing a Simple Web Server with Node.js
- V. *Debugging JavaScript*
 - a. Using Chrome's Developer Tools
 - b. Installing and Using the Vue DevTools
 - c. Debugging in Chrome
 - d. Lab 9: Manual Browser Testing
- VI. *Tooling*
 - a. Why have tooling?
 - b. npm and yarn
 - c. package.json
 - d. eslint and jshint
 - e. Lab 11: Automate Linting
 - f. transpilers
 - g. webpack
- VII. *Getting Started With Vue.js*
 - a. History of Vue.js
 - b. *What makes Vue different?*
 - i. *Vue vs React*
 - ii. *Vue vs Angular*
 - c. *Vue's Virtual DOM*
 - d. Vue Versions
 - i. What to expect in Vue 3
- VIII. *Basic Vue.js Features*
 - a. Understanding the Vue Instance
 - i. Instance properties and methods
 - b. Introduction to Vue's Directives
 - i. Arguments
 - ii. Modifiers
 - iii. Shorthands
 - c. Vue Lifecycle

- i. created
 - ii. mounted
 - iii. updated
 - iv. destroyed
 - v. Using Lifecycle Methods
 - d. Event Handling
 - i. Inline Event Handling
 - ii. Event Handling with Methods
 - iii. Event Modifiers
 - e. Vue Template Syntax
 - i. Using HTML
 - ii. Using JSX
 - iii. Using CreateElement
 - iv. Attributes
 - f. Loops and Lists
 - i. Using key
 - ii. Using v-for
 - 1. v-for with Arrays
 - 2. v-for with Objects
 - iii. Mutating arrays and objects
 - 1. Array and Object Change Detection
 - 2. The mutations Vue Can't Detect
 - g. Conditional Rendering
 - i. v-if vs. v-show
 - h. Using Filters
 - i. Formatting Currencies with Filters
 - ii. Formatting Dates with Filters
 - i. Computed Properties
 - i. Benefits of Computed Properties
 - ii. Filtering a List with a Computed Property
 - iii. Sorting a List with a Computed Property
 - iv. Using setters in computed properties
 - j. Creating custom watchers
 - i. Async operations with watchers
 - k. Binding HTML Classes
 - i. Displaying and Hiding an Element Conditionally
 - ii. Adding Styles Conditionally
 - iii. Binding Inline Styles
 - l. Outputting Raw HTML
 - m. Vue and Forms
 - i. Input Bindings
 - 1. value binding
 - 2. 2-way binding
 - ii. Creating a Form with Checkboxes
 - iii. Creating a Form with Radio Buttons
 - iv. Creating a Form with a Select Element
 - v. Input Modifiers
- IX. *All About Components*
 - a. Creating a Component
 - b. Organizing and Reusing Components
 - c. Event Handling with Components
 - d. Passing Data
 - i. Using Props

- ii. Using Events
 - iii. Using v-model
 - e. Reading a Child's State
 - f. Using Components in Your Own Components
 - g. Content Distribution with Slots
 - h. Single File Components with Webpack
 - i. Loading Your Components Asynchronously
- X. Mixins
 - a. Basics
 - b. Option Merging
 - c. Global Mixin
 - d. Custom Option Merge Strategies
- XI. Using Vuex
 - a. Understanding the Flux Pattern
 - b. The Store
 - i. Creating a Store
 - ii. store.state
 - iii. store.commit
 - c. Using Vuex State in Vue Components
 - d. Using mapState
 - e. Store getters
 - i. Using Getters
 - ii. Mapping Getters
 - f. Creating Mutation Types
 - g. Synchronous Mutations
 - i. Mutation handlers
 - ii. Passing data to store.commit
 - 1. Single value
 - 2. Object-style commit
 - iii. Committing Mutations in Components
 - h. Async Mutations with Actions
 - i. Actions vs. Mutations
 - ii. Dispatching Actions
 - iii. Composing Actions
 - i. Splitting the Store with Modules
 - i. Local State
 - ii. Namespacing
 - iii. Accessing Globals Assets from Namespaced Modules
 - iv. Registering Global Actions
 - v. Binding Helpers
 - vi. Dynamic Registration
 - vii. Reusing Modules
 - j. Plugins
- XII. *AJAX and Vue*
 - a. Sending Basic AJAX Requests with Axios
 - b. Validating User Data before Sending It
 - c. Recovering from an Error during a Request
 - d. Creating a REST Client (and Server!)
 - e. Implementing Infinite Scrolling
 - f. Processing a Request before Sending It Out
 - g. Preventing XSRF Attacks on Your App
- XIII. *Single-Page Applications*
 - a. Creating an SPA with Vue-Router
 - b. Fetching Data before Switching Route
 - c. Managing Errors for Your Routes

- d. Adding a Progress Bar to Load Pages
- e. Using Params
 - i. Reacting to Params Changes
- f. Having More Than One Router-View in Your Page
- g. Nested Routes
- h. Navigating Programmatically
- i. Using Named Dynamic Routes
- j. Adding Transitions between Your Routes
- k. How to Redirect to Another Route
- l. Passing Props to Route Components
- m. Route History Manipulation
- XIV. *Unit Testing and End-To-End Testing*
 - a. Unit Testing with Jest or Mocha
 - b. Vue Test Utils
 - c. Adding Karma to Your Workflow
 - d. Writing Testable Components
 - e. Testing Your Application State and Methods
 - f. Testing DOM Asynchronous Updates
 - g. End-to-end testing with TestCafe
 - h. Stubbing External API Calls with Sinon.JS
 - i. Measuring the Coverage of Your Code
- XV. *Organize + Automate + Deploy = Webpack*
 - a. Extracting Logic from Your Components to Keep the Code Tidy
 - b. Organizing Your Dependencies with Webpack
 - c. Using External Components in Your Webpack Project
 - d. Developing with Continuous Feedback with Hot Reloading
 - e. Running a Code Linter While Developing
 - f. Releasing Your Components to the Public
- XVI. *Asynchronous JavaScript (Time permitting)*
 - a. Multithreading
 - b. How the event loop works
 - c. Promises
 - d. Async and await
- XVII. *Transitions and Animations (Optional)*
 - a. Integrating with Third-Party CSS Animation Libraries
 - b. Adding Your Own Transition Classes
 - c. Animating with JavaScript Instead of CSS
 - d. Transitioning on the Initial Render
 - e. Transitioning Between Elements
 - f. Letting an Element Leave Before the Enter Phase in a Transition
 - g. Adding Entering and Leaving Transitions for Elements of a List
 - h. Transitioning Elements That Move in a List
 - i. Animating the State of Your Components
 - j. Dynamic Transitions